MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 11732354 LEAD, WINDAGE AND REFERENCE

This specification is approved for use by the U.S. Army Research and Development Command (ARRADCOM) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements and quality assurance provisions for the Circuit Card Assembly: 11732354 Lead, Windage and Reference which is the A4 component of the M21 Ballistic Computer Unit, 11732400.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-STD-45662	Calibration System Requirements

FSC 1220

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-810	Environmental Test Methods

DRAWINGS

ARRADCOM

11732354	Circuit Card Assembly: 11732354 Lead, Windage and Reference

Inspection Equipment

11750202	Automatic Card and Unit, Test Set
11750357	Adapter, Computer Board-A4
11821457	Fixture Shock, Vertical and Horizontal Plane
11821458	Shock Adapter, Plate Vib Horz-Vert Plane/Shock-Z
11821500	Vibration/Shock Test Fixture-Computer Unit FAT
11821520	Automatic Test Assembly Central Processor Unit
11821854	Test Fixture Auto Card, Test Set, Computer FAT

Packaging Data Sheet

P11732354	Packaging of Circuit Card Assembly: 11732354

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

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3. REQUIREMENTS

3.1 Fabrication. The lead, windage and reference circuit card assembly, herein referred to as the assembly, shall be manufactured in accordance with Drawing 11732354 and drawings pertaining thereto and when assembled, shall meet the requirements of this specification. (See 4.5.1)

3.1.1 Signals. The assembly shall provide the following signals:

	Signal Designation	Signal Nomenclature
a.	AC REF 1	AC reference one
b.	AC REF 2	AC reference two
c.	+10V	Positive dc reference
d.	-10V	Negative dc reference
e.	(Omega - K_W V_W)T_F	Time of flight multiplier signal
f.	MR1	Master reference one
g.	MR2	Master reference two

3.1.2 General specifications. The following provisions of MIL-F-13926 apply: (See 4.5.1)

a. Order of precedence
b. Dimensions and tolerance
c. Part identification and marking
d. Inorganic protective surface finishes
e. Workmanship

3.1.3 Ambient conditions. Standard ambient conditions shall be as follows:

a.	Temperature	73 $\pm$ 18°F
b.	Relative humidity	50 percent $\pm$ 30 percent
c.	Atmospheric pressure	28.5 + 2.0 $-$ 3.0 in. Hg.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample units for first article inspection and approval (see 4.4 and 6.2).

3.3 Performance. Unless otherwise specified, the assembly shall meet the performance requirements specified herein under standard ambient conditions of 3.1.3.

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TABLE I. Loads, power and logical sources.

Item	Conditions	Characteristics	Connections
1.0	Loads	As specified below 1/	Connected between following pins and P1-13A:
1.1	Resistor	360 ohms	P1-14A
1.2	Resistor	360 ohms	P1-16A
1.3	Resistor and Capacitor	1000 ohms in parallel with 6.8 microfarads (mfd)	P1-4B Negative capacitor terminal to P1-13A
1.4	Resistor and Capacitor	1000 ohms in parallel with 6.8 mfd	P1-25B Positive capacitor terminal to P1-13A
1.5	Resistor and Diode	1500 ohms in series with 1N3600 diode	P1-24A Anode to P1-13A
1.6	Resistor and Diode	1500 ohms in series with 1N3600 diode	P1-25A Anode to P1-13A
2.0	Power Sources		Applied between following pins and P1-13A (rtn):
2.1	15.000 vdc (+15V)	Tolerance: ± 0.150 vdc Ripple: 20 mv p-p max	P1-6A
2.2	-15.000 vdc (-15V)	Tolerance: ± 0.150 vdc Ripple: 20 mv p-p max	P1-21A
2.3	5.000 vdc (+5V)	Tolerance: ± 0.150 vdc Ripple: 20 mv p-p max	P1-15A
3.0	Logical Sources	As specified below 2/	Applied between following pins and P1-13A (rtn):
3.1	AS1	(Logical 1) 0.70 ± 0.50 vdc in series with 100 ohms (Logical 0) - 15.00 ± 0.50 vdc in series with 5IK ohms	P1-1A

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TABLE I. Loads, power and logical sources. - Continued

Item	Conditions	Characteristics	Connections
3.2	AS2	Same characteristics as item 3.1	P1-1B
3.3	AS3	Same characteristics as item 3.1	P1-2B
3.4	AS4	Same characteristics as item 3.1	P1-2A
3.5	AS5	Logical 0 of item 3.1	P1-3A
3.6	AS6	(Logical 1) 0.00 ± 0.50 vdc in series with 20K ohms (Logical 0) -15.00 ± 0.50 vdc in series with 1000 ohms	P1-12A
3.7	STAB	(Logical 1) 24 ± 2 vdc in series with 100 ohms (Logical 0) open, no connection	P1-10A
3.8	LL (LEADLOCK 2)	(Logical 1) 24 ± 2 vdc in series with 1000 ohms (Logical 0) open, no connection	P1-11A
3.9	MT	(Logical 1) 15.00 ± 0.50 vdc in series with 1000 ohms (Logical 0) -15.00 ± 0.50 vdc in series with 20K ohms	P1-10B
3.10	BIT	(Logical 1) 13.00 ± 0.50 vdc in series with 510 ohms (Logical 0) -14.50 ± 0.50 vdc in series with 10K ohms	P1-6B
3.11	$\overline{\text{BIT}}$ (BIT BAR)	(Logical 1) 13.50 ± 0.50 vdc in series with 510 ohms (Logical 0) -14.00 ± 0.50 vdc in series with 10K ohms Logical complement of item 3.10	P1-9A

1/ a. All resistors of table I are 1/4 watt, ± 5% tolerance.
b. Capacitors are solid tantalum, ± 10% tolerance with 35 vdc working voltage rating.

2/ Logical terms have a value of either one (1) or zero (0) as defined by voltage levels in characteristics column.

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3.3.1 Reference signal outputs. With loads (items 1.1 through 1.6 of table I) connected and power (items 2.1 and 2.2 of table I) applied, the following requirements shall be met after a stabilization period of not less than 30 seconds. Reference signals shall be measured with respect to P1-13A.

3.3.1.1 AC REF 1 output. The output V_{14A} at P1-14A shall be a sine wave with the following characteristics:

Item	Temperature	Frequency (Hz)	Amplitude (vrms)
1	Standard ambient	400 ± 1	5.000 ± 0.005
2	-25°F	398 ± 2.1	5.000 ± 0.005
3	140°F	402 ± 2	5.000 ± 0.005

This signal shall appear also at P1-78 and P1-15A. Unless specified otherwise, positive (+) magnitudes shall be represented by phase 1 signals which are in phase ± 1.0° with V_{14A} and negative (-) magnitudes shall be represented by phase 2 signals which are 180° ± 1° out of phase with V_{14A}. All amplitudes shall be measured as the root mean square (rms) value of the in-phase component of the fundamental where MR1 at P1-24A is the phase reference. (See 4.6.2.1)

3.3.1.2 AC REF 2 output. The output V_{16A} at P1-16A shall be described by the following transfer function:

$$V_{16A} = - 1.0000 * V_{14A} \quad \text{vrms}$$

The tolerance of V_{16A} shall be as follows:

Item	Temperature	Tolerance (vrms)
1	Standard ambient	± 0.005
2	-25°F	± 0.015
3	140°F	± 0.010

(See 4.6.2.1)

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3.3.1.3 +10V output. The output V_{4B} at P1-4B shall be described by the following transfer function:

$$V_{4B} = 2.000 * V_{14A}$$ volts direct current (vdc)

The tolerance of V_{4B} shall be as follows:

Item	Temperature	Tolerance (vdc)
1	Standard ambient	$\pm$ 0.006
2	-25°F	$\pm$ 0.045
3	140°F	$\pm$ 0.030

Ripple shall be not greater than 5 millivolts (mv) peak-to-peak (p-p). (See 4.6.2.1)

3.3.1.4 -10V output. The output V_{25A} at P1-25A shall be described by the following transfer function:

$$V_{25A} = - 1.000 * V_{4B}$$ vdc

The tolerance of V_{25A} shall be as follows:

Item	Temperature	Tolerance (vdc)
1	Standard ambient	$\pm$ 0.010
2	-25°F	$\pm$ 0.025
3	140°F	$\pm$ 0.014

(see 4.6.2.1)

3.3.1.5 MR1 and MR2 outputs. The MR1 output at pin P1-24A and MR2 output at pin P1-25A shall be as specified in figure 1. (See 4.6.2.1)

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TABLE II. $(Omega - K_W V_W)T_F$ inputs.

Item	Input Signal 1/	Pin Location	Signal Limits
1	V_{5B}	P1-5B	0.0 to 6.5 vrms phase 1
2	V_{4A}	P1-4A	0.0 to 7.5 vrms phase 1
3	V_{3B}	P1-3B	-10 to +10 vdc
4	V_{5A}	P1-5A	-3.75 to 3.75 vdc
5	V_{7A}	P1-7A	-3.75 to 3.75 vdc
6	V_{TACH}	See figure 2	-1.8 to 1.8 vdc in series with 15K ohms $\pm$ 0.05%

1/ Measured with respect to P1-13A.

3.3.2 $(Omega - K_W V_W)T_F$ output. With power (items 2.1, 2.2 and 2.3 of table I) applied, logical sources (items 3.1 through 3.11 of table I) applied and the inputs (items 1 through 6 of table II) applied, the nominal output of P1-24B shall be described by the following transfer function:

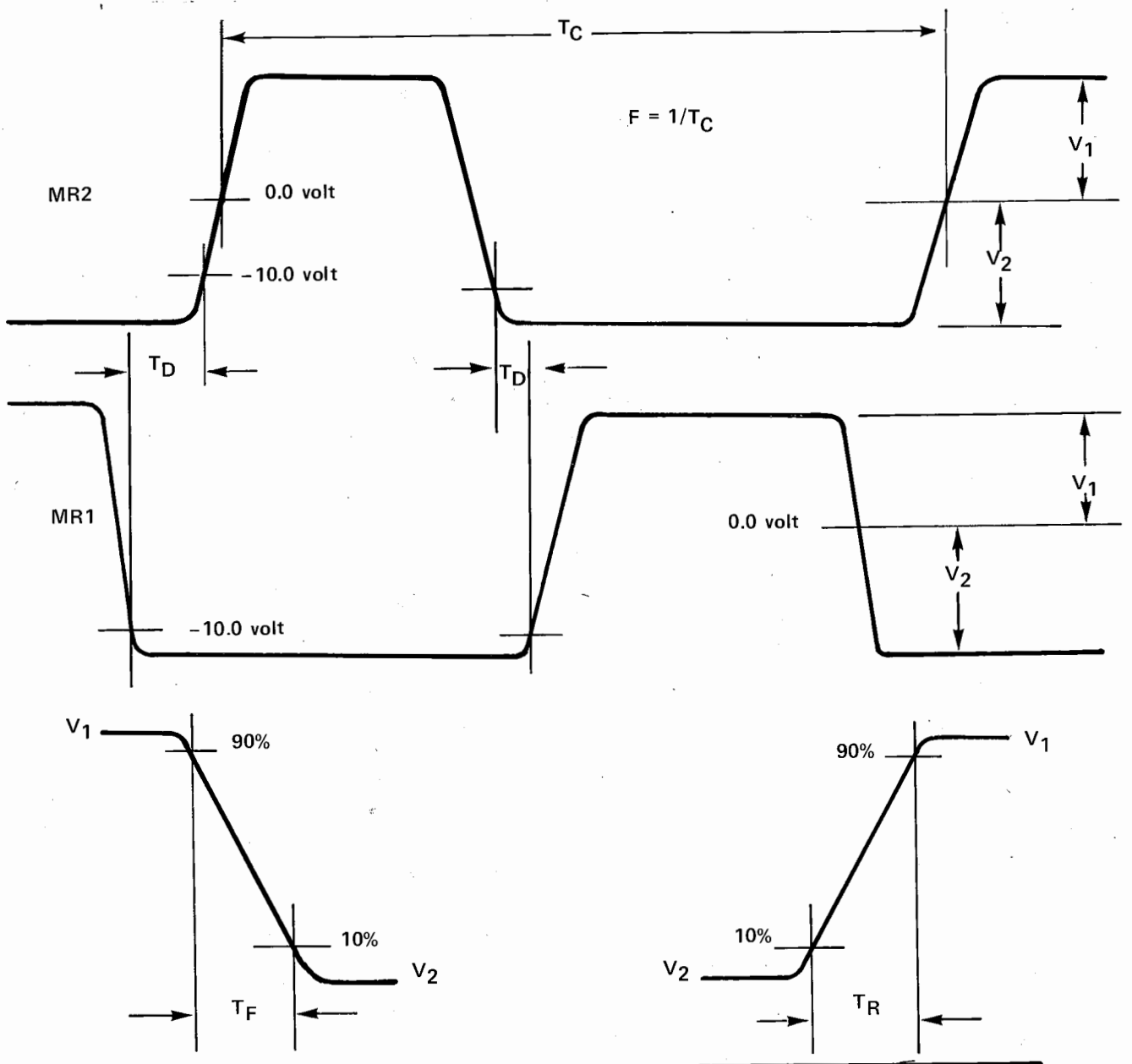
$$V_{24B} = \frac{(T * V_{Omega} + V_{24B}(0)) * V_{14A}}{5.000}$$

Where $V_{24B}(0)$ and quantities T and V_{Omega} are calculated from the following expressions:

a. $T = K_1 * V_{4A} + K_2 * V_{5B}$

b. $$V_{omega} = \frac{V_{TACH}}{1.20426} * 6.7912 * SW1 + 2.7165 * (V_{5A} - V_{7A}) * SW2 + K_3 * V_{4B} * SW3 + K_4 * V_{3B}$$

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	Standard ambient	140° F	-25° F
V_1	15 ± 0.50 volts	15 ± 0.50 volts	15 ± 0.50 volts
V_2	-15 ± 0.50 volts	-15 ± 0.50 volts	-15 ± 0.50 volts
T_D	1 to 30 microsecond	1 to 30 microsecond	1 to 35 microsecond
T_R	0.5 to 5 microsecond	0.5 to 4 microsecond	1 to 6 microsecond
T_F	0.5 to 4 microsecond	0.5 to 4 microsecond	0.5 to 6 microsecond
F	400 ± 1 Hz	402 ± 2 Hz	398 ± 2 Hz

FIGURE 1. Master reference waveforms.

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c. V_{24B} (0) is the output at P1-24B when:

1. $V_{3B} = 0.000 \pm 0.005$ vdc

2. $V_{14B} = -1.000 \pm 0.020$ vrms

3. LL = Logical 0

3.3.2.1 Definition of terms.

a. V_{TACH} is defined by figure 2.

b. The values of K_1, K_2, K_3 and K_4 shall be determined for each item from table III.

c. Terms SW1, SW2 and SW3 have the value of either 1 or 0 as defined by these logical expressions:

1. SW1 = LL AND $\overline{\text{BIT}}$ AND ($\overline{\text{MT AND STAB}}$)

2. SW2 = LL AND $\overline{\text{BIT}}$ AND MT AND STAB

3. SW3 = BIT AND $\overline{\text{LL}}$

3.3.2.2 Tolerance. The tolerance of the output shall be not greater than the following equation:

$$\Delta V_{24B} = 0.0020 * \left|V_{24B}\right| + 0.0093 * \left|T + 1\right| + \Delta T * \left|V_{Omega}\right| + \Delta V_{Omega} * \left|T\right|$$

Where Δ T and ΔV_{Omega} are calculated from the following expressions:

a. $\Delta T = \Delta K_1 * V_{4A} + \Delta K_2 * V_{5B}$

b. $\Delta V_{Omega} = (0.00271 * \left|V_{5A} - V_{7A}\right| + 0.022) * SW2 + (\Delta K_3 * V_{4B} + 0.002) * SW3 + \Delta K_4 * V_{3B} + 0.002 * t * LL$

3.3.2.2.1 Definition of terms.

a. Values for ΔK_1, ΔK_2, ΔK_3 and ΔK_4 shall be determined from table III.

b. t is the time in seconds after LL is set to Logical 1.

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3.3.2.3 Application limits. Application of the transfer function and the tolerance equation shall be valid within the following limits:

a. $|V_{24B}| < 5$ vrms

b. $V_{14B} < 6$ vrms

c. $t < 20$ seconds

d. $V_{24B}\ (0) < 0.012$ vrms

e. All inputs shall have achieved steady state for 9 ± 1 seconds before LL is set to logical 1. (See 4.6.2.2)

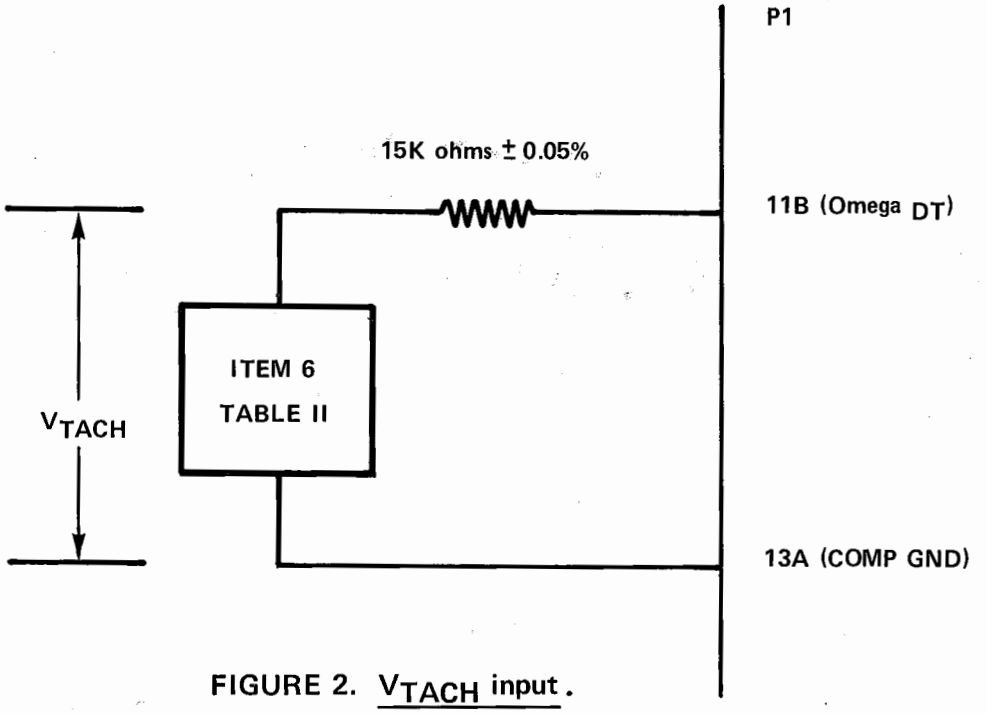


FIGURE 2. V_{TACH} input.

3.3.3 Logical input interaction. With the power and logical sources (items 2.1, 2.2, 3.8, 3.9, 3.10 and 3.11 of table I) applied, the voltages at P1 shall be as specified for each of the four items of table IV when the corresponding logical states are established. (See 4.6.2.3)

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TABLE III. (Omega - $K_w V_w$)T_F constants.

Item	cted ts 1/	K_1	K_2	K_3	K_4	ΔK_1	ΔK_2	ΔK_3	ΔK_4
1	AS1	-0.042000	-1.192000	0.068681	-0.014931	0.00013	0.00073	0.000427	0.000113
2	AS1,AS6	-0.042000	-1.192000	0.067670	-0.014931	0.00013	0.00073	0.000418	0.000113
3	AS2,AS6	-0.126366	-0.704149	0.112871	-0.057220	0.00000	0.00000	0.000544	0.000314
4	AS3,AS6	-0.218966	-0.416040	0.167743	-0.066560	0.00029	0.00053	0.000696	0.000348
5	AS4,AS6	-0.061726	-1.087000	0.066266	-0.009916	0.00019	0.00121	0.000414	0.000091

1/ Selected inputs shall be set to logical 1 (see Item 3.0 of table I)

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TABLE IV. Logical input interaction.

Item	Logical State 1/			Voltages at P-1 (vdc) 2/			
	BIT	MT	LL	P1-6B	P1-9A	P1-10B	P1-11A
1	1	0	1	13.00 ± 0.75	-14.00 ± 0.75	-14.60 ± 0.75	24.00 ± 2.50
2	1	1	0	13.00 ± 0.75	-14.00 ± 0.75	15.00 ± 0.75	-15.00 ± 0.75
3	0	0	1	-14.40 ± 0.75	13.50 ± 0.75	-14.60 ± 0.75	24.00 ± 2.50
4	0	1	0	-14.40 ± 0.75	13.50 ± 0.75	15.00 ± 0.75	-15.00 ± 0.75

1/ See item 3.0 of table I.
2/ Measured with respect to P1-13A.

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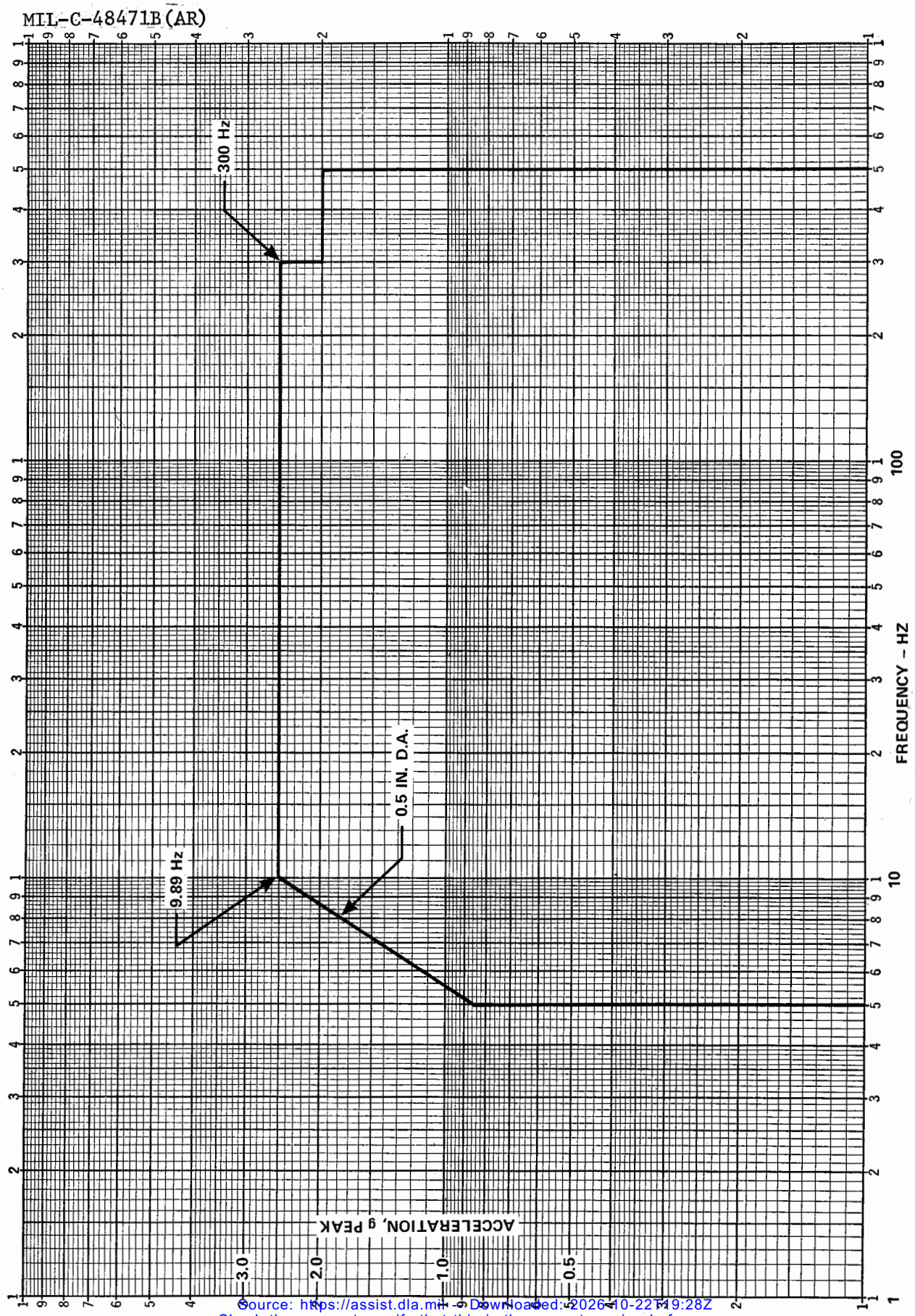


FIGURE 3. Vibration profile.

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3.4 Environmental.

3.4.1 Shock. The assembly shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to three half sine wave shock pulses of 40 ± 4 gravity unit (g) for a duration of 18 ± 3 milliseconds (msec) applied in each direction along three mutually perpendicular axes. In addition, the assembly shall be capable of withstanding three half sine wave shock pulses of 100.0 ± 10.0 g for a duration of 1.5 ± 0.2 msec applied in each direction along three mutually perpendicular axes. (See 4.6.3.1)

3.4.2 Vibration. The assembly shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to the vibration profile in figure 3. Duration of exposure shall be not less than 80 minutes in each of three mutually perpendicular axes. (See 4.6.3.2)

3.4.3 Operating temperature. The assembly shall be capable of operating as specified herein over the operating temperature range of -25°F to 140°F. (See 4.6.3.3, 4.6.3.4, and 4.6.3.5)

3.4.4 Storage temperature. The assembly shall be capable of operating as specified herein after exposure to storage temperatures ranging from -65°F to +160°F. (see 4.6.3.3, 4.6.3.4 and 4.6.3.5)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

1. First article inspection (See 4.4).
2. Quality conformance inspection (See 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the quality assurance provisions of MIL-F-13926 and the conditions of 3.1.3.

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4.4 First article. The requirement for first article approval and the designation of responsibility for first article inspection to either the Government or the contractor shall be specified in the contract.

4.4.1 Sample. The first article sample shall be three assemblies selected at random by the Government representative from the first fifteen production assemblies.

4.4.2 Inspection. The sample shall be subjected to all the inspections specified in tables V, VI and VII.

4.4.3 Failures. Failure of any assembly to meet any requirement shall be cause for refusal to grant first article approval. The Government reserves the right to terminate first article inspection upon any failure of an assembly to comply with any stated requirement.

4.4.4 Responsibility. The contractor, whether or not responsible, shall inspect the sample for conformance to all contractual requirements and shall submit a record of this inspection with the sample and certificates of conformance for materials. The Government reserves the right to witness inspections of the contractor.

TABLE V. Performance tests.

Item	Characteristic	Requirement	Test Procedures
101	Reference voltage supplies	3.3.1.1 through 3.3.1.5	4.6.2.1
102	(Omega - $K_W V_W$)T_F output	3.3.2	4.6.2.2
103	Logical input interaction	3.3.3	4.6.2.3

TABLE VI. General tests.

Item	Characteristic	Requirement	Test Procedures
301	Fabrication	3.1	Applicable drawings - Visual
302	General Specification	3.1.2	MIL-F-13926 - Visual

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TABLE VII. Environmental tests.

Item	Characteristic	Requirement	Test Procedures
303	Shock	3.4.1	4.6.3.1
304	Vibration	3.4.2	4.6.3.2
305	High Temperature	3.4.3 and 3.4.4	4.6.3.3
306	Low Temperature	3.4.3 and 3.4.4	4.6.3.4
307	Temperature Cycling	3.4.3 and 3.4.4	4.6.3.5

4.5 Quality conformance inspection.

4.5.1 Procurement conditions. The following inspection plans shall apply where the assembly is procured for use as a:

Use	Plan
a. Logistics spare assembly apart from the next higher assembly	A and C
b. Component assembly of the next higher assembly	B and C

4.5.2 Inspection plan A.

4.5.2.1 General sample and tests. One assembly, as a control sample, shall be selected at random by the Government representative from each 100 assemblies produced and shall be subjected to all the tests in table VI.

4.5.2.2 Environmental sample and tests. Three assemblies, as a control sample, shall be selected at random by the Government representative from each 50 assemblies produced or from each month's production, whichever occurs first. All the tests in table VII shall be applied separately to each assembly in the sample.

4.5.2.3 Acceptance. Where any one assembly of either sample fails to meet any specified requirement, the lot shall be rejected. Rejected lots shall be subject to the provisions of MIL-STD-105.

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4.5.3 Inspection plan B.

4.5.3.1 Sample and tests. The sample shall be selected by the Government representative in accordance with the provisions of MIL-STD-105. All the tests in table VI and only the test of item 307 in table VII shall be applied.

4.5.3.2 Acceptance. Acceptance and rejection shall be in accordance with MIL-STD-105.

4.5.4 Inspection plan C.

4.5.4.1 Sample and tests. Each assembly in every lot shall be subjected to all the tests in table V and shall be examined visually for completeness, improper assembly and evidence of poor workmanship.

4.5.4.2 Acceptance. Where any one assembly fails to meet any specified requirement, the defective assembly shall be removed from the lot and resubmitted only after all defects have been corrected.

4.5.5 Inspection equipment. Unless otherwise specified in the contract, the contractor shall supply, maintain and calibrate inspection equipment in accordance with the provisions of MIL-I-45607 and MIL-STD-45662.

4.5.6 Packaging inspection. The sampling and inspection of the preservation-packaging, packing and container marking shall be in accordance with the provisions of packaging data sheet P11732354.

4.6 Methods of inspection.

4.6.1 Test equipment. Unless the assembly is procured apart from the next higher assembly, the test equipment in table VIII shall be used where specified to perform the required tests.

TABLE VIII. Test equipment.

Item	Part Number	Nomenclature
1	11750202	Automatic Card and Unit, Test Set
2	11750357	Adapter, Computer Board - A4
3	11821457	Fixture Shock, Vertical and Horizontal Plane
4	11821458	Shock Adapter, Plate Vib Horz-Vert Plane/Shock-Z
5	11821500	Vibration/Shock Test Fixture - Computer Unit FAT
6	11821520	Automatic Test Assembly Central Processor Unit
7	11821854	Test Fixture Auto Card, Test Set, Computer, FAT

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4.6.2 Performance tests.

4.6.2.1 Reference voltage supplies. Use items 1, 2 and 6 of table VIII or suitable equipment to verify the requirements of 3.3.1.1 through 3.3.1.5.

4.6.2.2 (Omega - $K_W V_W$)T_F output. Use items 1, 2 and 6 of table VIII or suitable equipment to verify the requirements of 3.3.2.

4.6.2.3 Logical input interaction. Use items 1, 2 and 6 of table VIII or suitable equipment to verify the requirements of 3.3.3.

4.6.3 Environmental tests.

4.6.3.1 Shock.

4.6.3.1.1 Basic design. Use items 3, 4 and 5 of table VIII or mount the assembly in a suitable shock fixture and subject it to the shock test specified in MIL-STD-810, method 516.2, procedure I, figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 40 $\pm$ 4 g with a time duration of 18 $\pm$ 3 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the assembly to the performance tests of table V.

4.6.3.1.2 High intensity. Use items 3, 4 and 5 of table VIII or mount the assembly in a suitable shock fixture and subject it to the shock test specified in MIL-STD-810, method 516.2, procedure IV, figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 100 $\pm$ 10.0 g with a time duration of 1.5 $\pm$ 0.2 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the assembly to the performance tests of table V.

4.6.3.2 Vibration. Vibration A as specified in 4.6.3.2.1 shall be conducted only for first article inspection. Vibration B as specified in 4.6.3.2.2 shall be conducted only for quality conformance inspection.

4.6.3.2.1 Vibration A. Use items 4 and 5 of table VIII or mount the assembly in a suitable vibration fixture and subject it to the vibration test specified in MIL-STD-810, method 514.2, procedure VIII, except that the test level shall be the vibration curve shown in figure 3 and duration of exposure shall be not less than 80 minutes per axis. At the conclusion of this test, subject the assembly to the performance tests of table V.

4.6.3.2.2 Vibration B. This test shall be conducted as in 4.6.3.2.1 except that the period of vibration shall be not less than 15 minutes in each axis.

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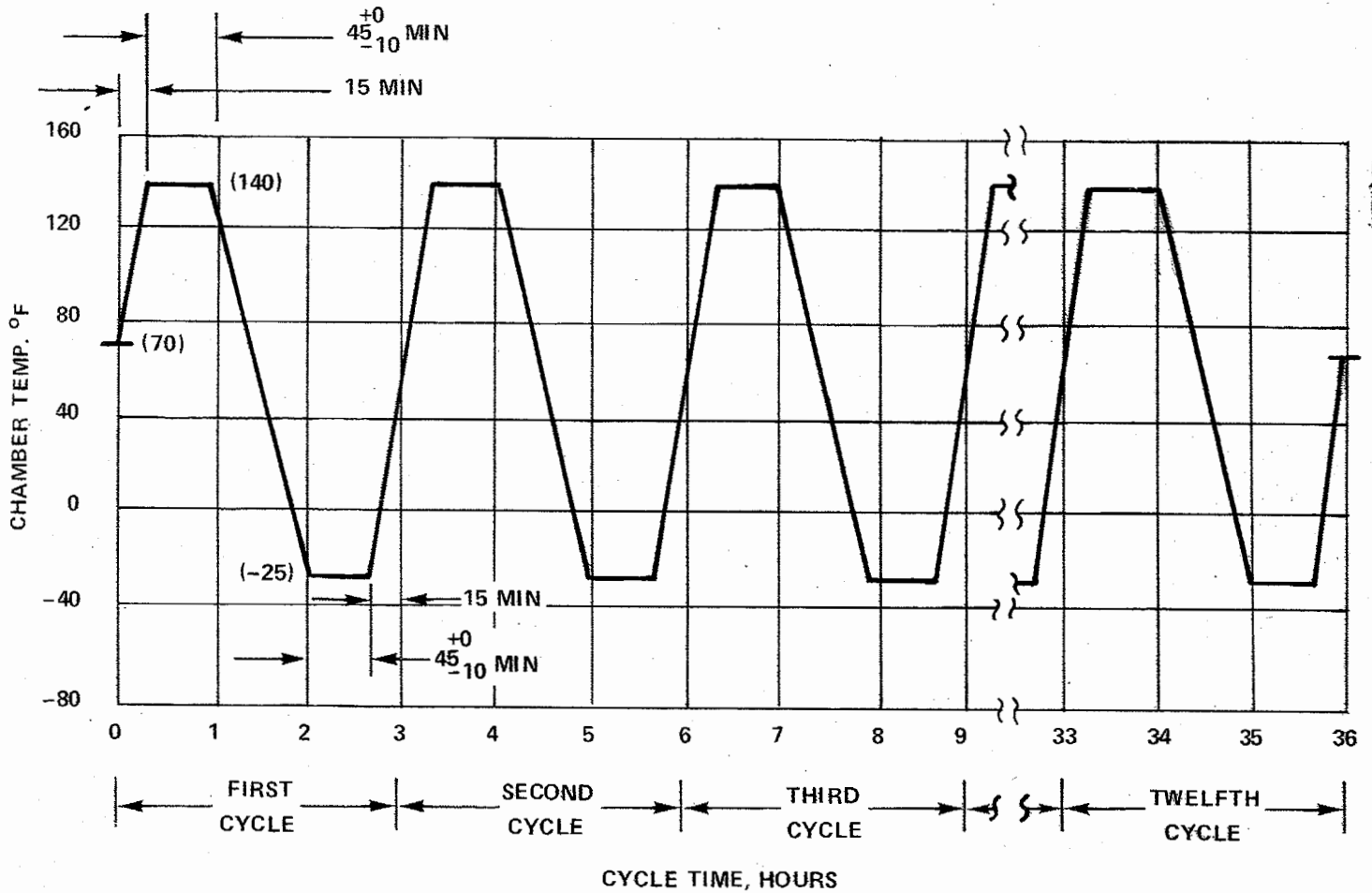


FIGURE 4. Temperature cycling.

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4.6.3.3 High temperature. Use item 7 of table VIII or suitable equipment and subject the assembly to the high temperature test specified in MIL-STD-810, method 501.1, procedure I. The highest operating temperature shall be 140°F. The performance tests of table V shall be applied.

4.6.3.4 Low temperature. Use item 7 of table VIII or suitable equipment and subject the assembly to the low temperature test specified in MIL-STD-810, method 502.1, procedure I. The storage temperature shall be -65°F. The lowest operating temperature shall be -25°F. The performance tests of table V shall be applied.

4.6.3.5 Temperature cycling. With power and loads of table I applied, subject the assembly to the temperature profile specified in figure 4. At the conclusion of this test, subject the assembly to the performance tests of table V.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with packaging data sheet P11732354. The level of protection shall be as specified in the procurement document. (See 4.5.6)

6. NOTES

6.1 Intended use. The Lead Windage and Reference circuit card assembly is used in the M21 Ballistic Computer Unit.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Special tests and examination as required, the type of inspection required, and the activity responsible for testing and its facility.

c. Applicable packaging data sheet number (see 5.1).

d. Selection of applicable levels of preservation, packaging and packing.

e. Applicable stock number.

f. First article inspection requirements.

6.3 Previous revisions of MIL-C-48471B was for the US Army Armament Research and Development Command's internal use only, and were not forwarded to the Naval Publications and Form Center for publication.

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